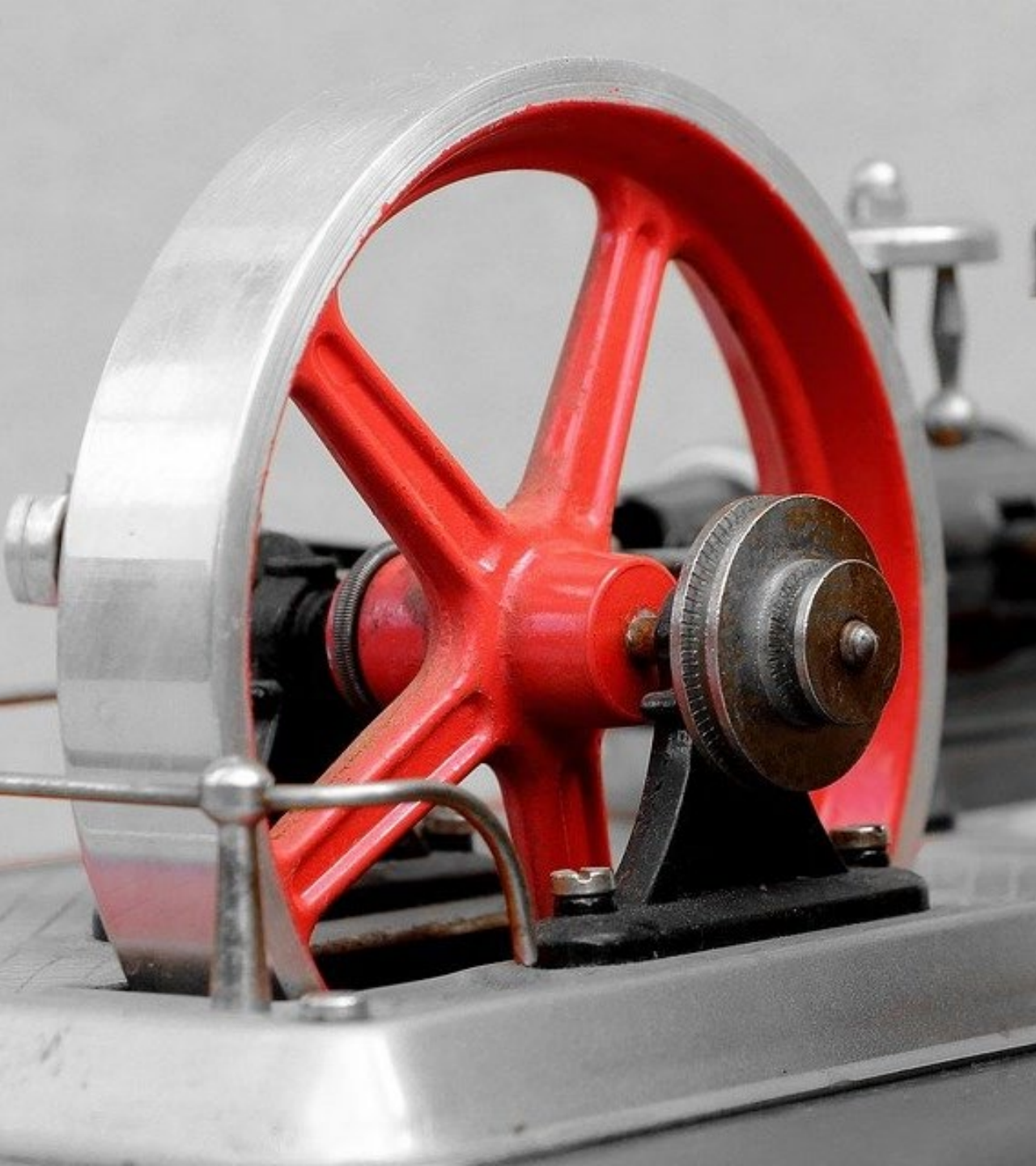
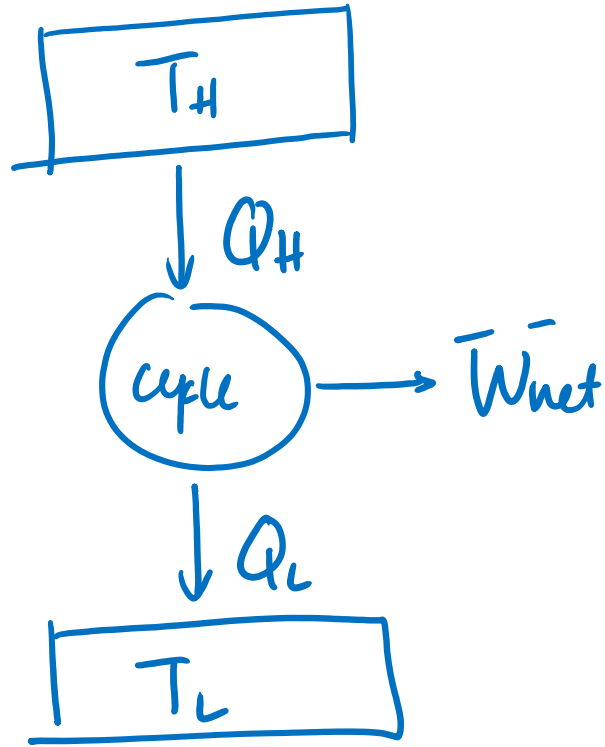




ME 300:
Engineering Thermodynamics

Consequences of the Kelvin-Planck Statement

Kelvin-Planck Statement (1B)

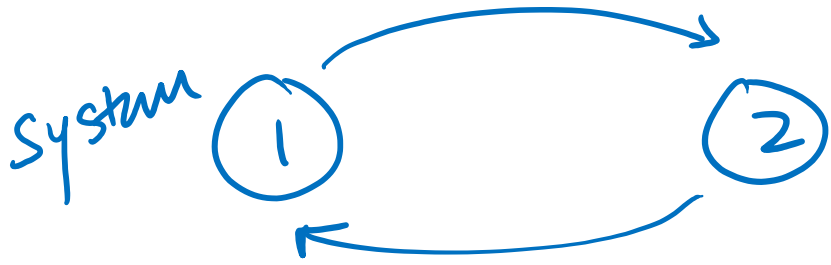


It is impossible to construct a cyclically-operating device for which the sole effect is the exchange of heat from a single reservoir and creation of equivalent amount of work.

Kelvin-Planck Statement (1B)

- work is "valuable" $\bar{W} \rightarrow Q$ friction, $Q \nrightarrow \bar{W}$ continuously
- Q_L is finite, $Q_H > \bar{W}_{net}$, have to reject heat
- maximize η_{th} , reduce "irreversibilities" - loss mechanisms
 - reversible cycle has highest η_{th}
- all reversible heat engines have the same η_{th} if they are operated between the same T_H, T_L
 - create a absolute temperature scale.

Reversible processes



cycle: system returns to its original State.

$\Rightarrow$ if all processes in cycle are reversible, then
system and its surroundings return to their
original State.

$\Rightarrow$ maximizes η_{th}

Temperature scale

→ 0th law: thermal eq.

$$\rightarrow \eta_{th, rev} = f(T_H, T_L)$$

$$\eta_{th} = 1 - \frac{Q_L}{Q_H} \Rightarrow \eta_{th, rev} = 1 - \frac{T_L}{T_H} \quad (\text{max})$$

$$\boxed{\frac{T_L}{T_H} = \frac{Q_L}{Q_H}}$$

if reversible cycle